Willian T. Lunardi

Principal Researcher at TII

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Education

Ph.D. in Computer Science 2016 – 2020
University of Luxembourg Luxembourg

M.Sc. in Computer Science 2014 – 2016
Pontifical Catholic University of Rio Grande do Sul RS, Brazil

B.Sc. in Computer Science 2010 – 2014
University of Passo Fundo RS, Brazil

Professional Experience

Principal Researcher Apr 2023 – Present
Technology Innovation Institute Abu Dhabi, UAE

- Lead R&D and post-training of LLM agents for autonomous software engineering and terminal tasks.
- Develop training strategies targeting generalizable reasoning, planning, tool use, and long-horizon execution.
- Design and oversee supervised fine-tuning of LLMs using synthetic tasks and agent trajectories.
- Lead researchers and engineers in agent development and evaluation.

Senior Researcher Jun 2021 – Apr 2023
Technology Innovation Institute Abu Dhabi, UAE

- Developed contrastive learning and adversarial training methods for OOD and anomaly detection.
- Designed hyperspherical representation models for robust anomaly detection.
- Conducted empirical studies and contributed to peer-reviewed research with cross-disciplinary teams.

Research Associate Jun 2020 – May 2021
University of Luxembourg Luxembourg

- Researched neural combinatorial optimization and developed predictive maintenance models for industry.
- Co-supervised PhD and MSc students on model development, experimentation, and publication.
- Led industry collaborations, translating operational requirements into research projects.

Doctoral Researcher Jul 2016 – Jun 2020
University of Luxembourg Luxembourg

- Developed mathematical models and optimization methods for industrial scheduling problems.
- Collaborated with academic and European industry partners on scheduling research.

Technical Skills

LLM & Agentic AI: Software engineering agents, terminal agents, tool calling, context management.
Agent Evaluation: Terminal-Bench 2.1, agent benchmarking, trajectory analysis, failure analysis.
LLM Post-Training: Supervised fine-tuning (SFT), synthetic data generation, training-trajectory pipelines.
Deep Learning: PyTorch, Hugging Face Transformers, PyTorch Lightning, PyTorch Geometric.
Infrastructure: Kubernetes, Docker, sandboxed execution, Linux, Git.
Programming: Python, C++, Bash.

Honors & Awards

Best Paper Award — ECML PKDD, 2025.
Excellent Thesis Award — University of Luxembourg, 2026.

Additional Information

Dual Citizenship: Brazil and Italy.
Languages: Portuguese (native), English (fluent), Italian (basic).
Relocation: Open to relocation globally.

Publications

[†] *Equal contribution. Links open publisher records or open manuscripts.*

Published and accepted papers

[2026] **W. T. Lunardi**, S. Shrestha, and M. Andreoni, "Learning Hyperspherical Time–Frequency Representations for Time-Series Out-of-Distribution Detection," in *International Joint Conference on Artificial Intelligence (IJCAI)*, 2026, pp. 2888–2896 [Paper]

[2026] M. Hull, H. Yang, P. Mehta, M. Phute, A. Cho, H. Wang, M. Lau, W. Lee, **W. Lunardi**, M. Andreoni, and D. H. Chau, "ComplicitSplat: Downstream Models are Vulnerable to Blackbox Attacks by 3D Gaussian Splat Camouflages," in *British Machine Vision Conference (BMVC)*, Accepted, 2026 [Paper]

[2026] H. Wang, M. Lau, A. Helbling, M. Hull, S. Peng, M. Phute, M. Andreoni, **W. T. Lunardi**, D. H. Chau, and W. Lee, "ARMOR: Manifold-Oriented Training for Adversarially Robust Aerial Object Detection under Data Scarcity," in *IEEE International Conference on Big Data (BigData)*, Accepted, short paper, 2026 [Paper]

[2026] M. Phute, M. Hull, H. Wang, A. Helbling, S. Peng, **W. Lunardi**, M. Andreoni, W. Lee, and D. H. Chau, "Differentiable Rendering Powered End-to-End Adversarial Attack Evaluation," in *Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD)*, 2026, pp. 107–120 [Paper]

[2026] M. Gallacher, C. A. Boano, A. S. M. S. Pillai, U. Roedig, **W. Lunardi**, and M. Baddeley, "SPIDER: Lightweight Speaker Identification on Resource-Constrained Embedded Devices," in *Proceedings of the ACM/IEEE International Conference on Embedded Artificial Intelligence and Sensing Systems (SenSys)*, 2026, pp. 1316–1329 [Paper]

[2026] E. de Matos, H. Alameri, **W. T. Lunardi**, M. Andreoni, and E. Viegas, "Toward an Intrusion Detection System for a Virtualization Framework in Edge Computing," in *IEEE Wireless Communications and Networking Conference (WCNC)*, 2026, pp. 1–7 [Paper]

[2025] M. Hull, H. Wang, M. Lau, A. Helbling, M. Phute, C. Zhang, Z. Kira, **W. Lunardi**, M. Andreoni, W. Lee, and D. H. Chau, "RenderBender: A Survey on Adversarial Attacks Using Differentiable Rendering," in *International Joint Conference on Artificial Intelligence (IJCAI)*, 2025, pp. 10 473–10 480 [Paper]

[2025] **W. T. Lunardi**, A. Banabila, D. Herzalla, and M. Andreoni, "Contrastive Representation Modeling for Anomaly Detection," in *European Conference on Artificial Intelligence (ECAI)*, 2025, pp. 2594–2601 [Paper]

[2025] D. Herzalla[†], **W. T. Lunardi**[†], and M. Andreoni, "Graph Neural Networks for Jamming Source Localization," in *Machine Learning and Knowledge Discovery in Databases (ECML PKDD)*, **Best Paper Award**, 2025, pp. 331–348 [Paper]

[2025] X. Tan, J. Sundar, R. Bruzzone, P. Ong, **W. T. Lunardi**, M. Andreoni, P. Tabuada, and A. D. Ames, "Secure Safety Filter: Towards Safe Flight Control under Sensor Attacks," in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025, pp. 195–202 [Paper]

[2025] G. Gebrehans, N. Ilyas, K. Eledlebi, **W. T. Lunardi**, M. Andreoni, C. Y. Yeun, and E. Damiani, "Generative Adversarial Networks for Dynamic Malware Behavior: A Comprehensive Review, Categorization, and Analysis," *IEEE Transactions on Artificial Intelligence*, vol. 6, no. 8, pp. 1955–1976, 2025 [Paper]

[2025] M. Hull, H. Yang, P. Mehta, M. Phute, A. Cho, H. Wang, M. Lau, W. Lee, **W. T. Lunardi**, M. Andreoni, and P. Chau, "3D Gaussian Splat Vulnerabilities," in *CVPR Workshop on Neural Fields Beyond Conventional Cameras (NFBCC)*, 2025 [Paper]

[2025] H. Wang, Z. Yang, S. Park, Y. Yang, S. Kim, **W. T. Lunardi**, M. Andreoni, T. Kim, and W. Lee, "SoundBoost: Effective RCA and Attack Detection for UAV via Acoustic Side-Channel," in *IEEE/IFIP International Conference on Dependable Systems and Networks (DSN)*, 2025, pp. 289–302 [Paper]

[2025] G. Gebrehans, **W. T. Lunardi**, and E. Damiani, "Boosting GAN Performance: Feature Transformation for Heavy-Tailed Malware Data Generation," in *Security and Privacy in Communication Networks (SecureComm 2024)*, 2025, pp. 258–271 [Paper]

[2024] M. Lau, H. Wang, A. Helbling, M. Hull, S. Peng, M. Andreoni, **W. T. Lunardi**, and W. Lee, "Non-Robust Features are Not Always Useful in One-Class Classification," in *CVPR Workshop on Visual Anomaly and Novelty Detection (VAND)*, 2024 [Paper]

[2024] M. Andreoni, **W. T. Lunardi**, G. Lawton, and S. Thakkar, "Enhancing Autonomous System Security and Resilience with Generative AI: A Comprehensive Survey," *IEEE Access*, 2024 [Paper]

[2024] S. Alseiari, **W. T. Lunardi**, and M. Andreoni, "Lightweight Fault Detection in UAVs: A Machine Learning Approach with Dynamic Time Windows," in *EWSN Workshop on Security and Privacy in Connected Embedded Systems (SPICES)*, 2024 [Paper]

[2024] A. S. Ali, **W. T. Lunardi**, G. Singh, L. Bariah, M. Baddeley, M. A. Lopez, J.-P. Giacalone, and S. Muhaidat, "RF Jamming Dataset: A Wireless Spectral Scan Approach for Malicious Interference Detection," *IEEE Communications Magazine*, vol. 62, no. 11, pp. 114–120, 2024 [Paper]

[2024] E. de Matos, **W. T. Lunardi**, J. Ukkonen, and T. Salminen, "An seL4-based Trusted Execution Environment on RISC-V," in *International Wireless Communications and Mobile Computing (IWCMC)*, 2024, pp. 712–717 [Paper]

[2023] A. R. Ba Nabila, E. K. Viegas, A. AlMahmoud, and **W. T. Lunardi**, "A Generative Adversarial Network-based Attack for Audio-based Condition Monitoring Systems," in *IEEE Consumer Communications & Networking Conference*, 2023, pp. 275–280 [Paper]

[2023] **W. T. Lunardi**, M. A. Lopez, and J.-P. Giacalone, "ARCADE: Adversarially Regularized Convolutional Autoencoder for Network Anomaly Detection," *IEEE Transactions on Network and Service Management*, vol. 20, no. 2, pp. 1305–1318, 2023 [Paper]

[2023] M. Gallacher, C. A. Boano, M. S. A. Sankar, U. Roedig, **W. T. Lunardi**, and M. Baddeley, "Towards Speaker Identification on Resource-Constrained Embedded Devices," in *ACM Conference on Embedded Networked Sensor Systems*, 2023 [Paper]

[2023] D. Herzalla, **W. T. Lunardi**, and M. Andreoni, "TII-SSRC-23 Dataset: Typological Exploration of Diverse Traffic Patterns for Intrusion Detection," *IEEE Access*, vol. 11, pp. 118 577–118 594, 2023 [Paper]

[2022] A. S. Ali, **W. T. Lunardi**, L. Bariah, M. Baddeley, M. A. Lopez, J.-P. Giacalone, and S. Muhaidat, "Deep Reinforcement Learning Based Anti-Jamming Using Clear Channel Assessment Information in a Cognitive Radio Environment," in *IEEE International Conference on Advanced Communication Technologies and Networking*, 2022 [Paper]

[2022] A. S. Ali, M. Baddeley, L. Bariah, M. A. Lopez, **W. T. Lunardi**, J.-P. Giacalone, and S. Muhaidat, "JamRF: Performance Analysis, Evaluation, and Implementation of RF Jamming Over Wi-Fi," *IEEE Access*, 2022 [Paper]

[2022] A. S. Ali, M. Baddeley, L. Bariah, M. A. Lopez, **W. T. Lunardi**, J.-P. Giacalone, and S. Muhaidat, "Performance Analysis and Evaluation of RF Jamming in IoT Networks," in *IEEE Global Communications Conference*, 2022 [Paper]

[2021] **W. T. Lunardi**, E. G. Birgin, D. P. Ronconi, and H. Voos, "Metaheuristics for the Online Printing Shop Scheduling Problem," *European Journal of Operational Research*, 2021 [Paper]

[2021] M. A. Lopez, M. Baddeley, **W. T. Lunardi**, A. Pandey, and J.-P. Giacalone, "Towards Secure Wireless Mesh Networks for UAV Swarm Connectivity: Current Threats, Research, and Opportunities," in *IEEE International Conference on Distributed Computing in Sensor Systems*, 2021 [Paper]

[2020] **W. T. Lunardi**, E. G. Birgin, P. Laborie, D. P. Ronconi, and H. Voos, "Mixed Integer Linear Programming and Constraint Programming Models for the Online Printing Shop Scheduling Problem," *Computers & Operations Research*, 2020 [Paper]

[2019] **W. T. Lunardi**, H. Voos, and L. H. Cherri, "An Effective Hybrid Imperialist Competitive Algorithm and Tabu Search for an Extended Flexible Job Shop Scheduling Problem," in *ACM Symposium on Applied Computing*, 2019 [Paper]

[2019] H. de Faria Jr, **W. T. Lunardi**, and H. Voos, "A Parallel Multi-Population Biased Random-Key Genetic Algorithm for Electric Distribution Network Reconfiguration," in *Genetic and Evolutionary Computation Conference Companion (GECCO)*, 2019, pp. 281–282 [Paper]

[2018] **W. T. Lunardi**, L. H. Cherri, and H. Voos, "A Mathematical Model and a Firefly Algorithm for an Extended Flexible Job Shop Problem with Availability Constraints," in *Springer International Conference on Artificial Intelligence and Soft Computing*, 2018 [Paper]

[2018] **W. T. Lunardi** and H. Voos, "Comparative Study of Genetic and Discrete Firefly Algorithm for Combinatorial Optimization," in *ACM Annual Symposium on Applied Computing*, 2018 [Paper]

[2018] **W. T. Lunardi** and H. Voos, "An Extended Flexible Job Shop Scheduling Problem with Parallel Operations," *ACM SIGAPP Applied Computing Review*, 2018 [Paper]

[2018] **W. T. Lunardi**, H. Voos, and L. H. Cherri, "An Imperialist Competitive Algorithm for a Real-World Flexible Job Shop Scheduling Problem," in *IEEE International Conference on Emerging Technologies & Factory Automation*, 2018 [Paper]

[2016] **W. T. Lunardi**, L. Amaral, S. Marczak, F. Hessel, and H. Voos, "Automated Decision Support IoT Framework," in *IEEE International Conference on Emerging Technologies & Factory Automation*, 2016 [Paper]

[2016] L. A. Amaral, E. de Matos, R. T. Tiburski, F. Hessel, **W. T. Lunardi**, and S. Marczak, "Middleware Technology for IoT Systems: Challenges and Perspectives Toward 5G," in *Internet of Things (IoT) in 5G Mobile Technologies*. 2016 [Paper]

[2015] E. de Matos, L. A. Amaral, R. Tiburski, **W. T. Lunardi**, F. Hessel, and S. Marczak, "Context-Aware System for Information Services Provision in the Internet of Things," in *IEEE Conference on Emerging Technologies & Factory Automation*, 2015 [Paper]

[2015] **W. T. Lunardi**, E. de Matos, R. Tiburski, L. A. Amaral, S. Marczak, and F. Hessel, "Context-Based Search Engine for Industrial IoT: Discovery, Search, Selection, and Usage of Devices," in *IEEE Conference on Emerging Technologies & Factory Automation*, 2015 [Paper]

Preprints

[2026] M. Phute, A. Greenhalgh, M. Hull, H. Wang, A. Helbling, S. Peng, E. Faa, **W. Lunardi**, M. Andreoni, W. Lee, and D. H. Chau, "ALLUDE: A Unified Evaluation System for Configurable Attacks in Differentiable Environments," *arXiv preprint arXiv:2607.17077*, 2026 [Paper]

[2026] D. Herzalla, G. Singh, **W. T. Lunardi**, and M. Andreoni, "WiFiSpectralJam: A Large-Scale Open Wi-Fi Spectral Scan Dataset with Controlled RF Jamming," *arXiv preprint arXiv:2608.15728*, 2026 [Paper]